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PHOTOSYNTHESIS & LIGHT REACTIONS

Photosynthesis is the biochemical mechanism by which autotrophic plants convert electromagnetic radiation from sunlight into chemical bonds stored as carbohydrates.

1. Light-Dependent Reactions occur in Thylakoid membranes inside the chloroplast:
- Photons excite chlorophyll P680 within Photosystem II (PS II).
- Photolysis of Water: 2 H2O -> 4 H+ + 4 e- + O2. Oxygen gas is released as a vital byproduct.
- The electron transport chain pumps protons into the thylakoid lumen, generating a strong proton-motive force.
- ATP Synthase uses chemiosmotic proton return into the stroma to synthesize ATP from ADP and inorganic phosphate.
- High-energy electrons reduce NADP+ via ferredoxin and NADP+ reductase to generate NADPH.

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CALVIN-BENSON CYCLE (LIGHT-INDEPENDENT STROMA REACTIONS)

The second stage occurs in the chloroplast stroma and consumes ATP and NADPH produced during the light phase:
1. Carbon Fixation: The enzyme RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) fixes atmospheric CO_2 onto a 5-carbon RuBP molecule.
2. Reduction: ATP and NADPH convert 3-PGA into glyceraldehyde-3-phosphate (G3P).
3. RuBP Regeneration: Five out of every six G3P molecules are recycled using ATP to regenerate three RuBP acceptors.

Net balance for 1 molecule of glucose: $6\ CO_2 + 18\ ATP + 12\ NADPH \rightarrow C_6H_{12}O_6 + 18\ ADP + 12\ NADP^+$.